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CSE-476: Introduction to Natural Language Processing

Quiz 1

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Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

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1. In the BIO scheme, which tag marks tokens that are inside an entity span?
 - A. Beginning of an entity
 - B. Inside an entity
 - C. Outside any entity
 - D. Between two tokens
2. In sequence tagging, which model variant considers only the observed word sequence and ignores tag labels?
 - A. It uses only words (observations) and ignores labels
 - B. It is identical to an HMM
 - C. It models transitions between labels but does not consider the words
 - D. It models emissions $P(\text{word} \mid \text{label})$
3. In an HMM for NER, which probability term captures how likely one label follows the previous label?
 - A. $P(\text{label} \mid \text{previous label})$
 - B. $P(\text{label} \mid \text{word})$
 - C. $P(\text{word} \mid \text{previous word})$
 - D. $P(\text{word} \mid \text{label})$
4. Which option describes a task that assigns grammatical roles like subject and object based purely on syntax?
 - A. A task that identifies predicates in a sentence and labels roles of elements such as agent, theme, or location

- B.** A process that assigns grammatical roles like subject and object based on syntactic structure
 - C.** A method that classifies named entities and links them to a knowledge base
 - D.** A technique that labels words with their parts of speech and morphological features
5. Which option describes a compression scheme that replaces repeated character runs with a count and symbol?
- A.** A technique that compresses text by replacing consecutive repeated characters with a count and a symbol
 - B.** A tokenization method that iteratively merges the most frequent pair of symbols in a corpus
 - C.** A tokenization method that splits text based on whitespace and punctuation
 - D.** A character-level encoding scheme that assigns fixed-length binary codes to characters
6. Which option describes the task whose main goal is to assign a grammatical category to each token?
- A.** To label each token with a grammatical category
 - B.** To convert text into a sequence of symbols suitable for modeling
 - C.** To reduce vocabulary size to zero
 - D.** To assign probabilities to sentences
7. In English text, what is a major drawback of a tokenizer that completely ignores punctuation marks?
- A.** It produces too many tokens
 - B.** It does not work well for languages without explicit word boundaries
 - C.** It cannot tokenize English at all
 - D.** It ignores punctuation entirely
8. Which of the following is an example of a common sequence tagging task?
- A.** Part-of-speech tagging
 - B.** Named Entity Recognition
 - C.** Dependency parsing
 - D.** Sentiment analysis

9. Which option describes a drawback where the tagging scheme requires a very large number of distinct labels?
- A.** It cannot represent overlapping or nested entities
 - B.** It requires too many labels
 - C.** It does not support multiple entity types
 - D.** It ignores entity boundaries
10. When deciding between a fast but rough model and a slower but more accurate one, which trade-off is most relevant?
- A.** Speed vs accuracy
 - B.** Tokenized sequence length vs vocabulary size
 - C.** Precision vs recall
 - D.** Memory vs computation